THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour	Name: _____ Roll No.: _____	**Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The balanced equation for burning of magnesium ribbon in air is: 1

(a) $2\,Mg + O_2 \longrightarrow 2\,MgO_2$ **(b)** $Mg + O \longrightarrow MgO$ **(c)** $Mg + O_2 \longrightarrow MgO$ **(d)** $2\,Mg + O_2 \longrightarrow 2\,MgO$

2. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 7 **(b)** 10 **(c)** 1 **(d)** 4

3. Which of the following pairs will give a displacement reaction? 1

(a) NaCl solution and copper metal **(b)** $Fe + CuSO_4$ solution

(c) $MgCl_2$ solution and aluminium metal **(d)** $FeSO_4$ solution and silver metal

4. What would be the electron dot structure of carbon dioxide (CO_2)? The number of shared electron pairs in it is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 2

5. The kidneys in human beings are a part of the system for: 1

(a) respiration **(b)** transportation of oxygen

(c) excretion **(d)** nutrition

6. A part of the neuron where information is acquired is the: 1

(a) cell body **(b)** axon **(c)** dendrite **(d)** axon end

7. Binary fission in Amoeba differs from that in Leishmania because in Leishmania the division occurs 1

(a) only after sporulation

(b) at a definite plane related to the whip-like flagellum

(c) in any plane

(d) in the nucleus alone

8. Who is called the Father of Genetics? 1

(a) Charles Darwin **(b)** Gregor Johann Mendel

(c) Robert Hooke **(d)** Louis Pasteur

9. The angle of incidence is always equal to the angle of reflection. This is the 1

(a) law of refraction **(b)** Snell's law

(c) second law of reflection **(d)** first law of reflection

10. The image in the human eye is formed on the 1

(a) cornea **(b)** iris **(c)** retina **(d)** pupil

11. The SI unit of electric charge is 1

(a) ohm **(b)** ampere **(c)** coulomb **(d)** volt

12. The magnetic field lines outside a bar magnet are directed 1

(a) along the magnet only **(b)** from the north pole to the south pole

(c) in no definite direction **(d)** from the south pole to the north pole

13. An ecosystem consists of 1

(a) only non-living things **(b)** only animals

(c) biotic and abiotic components **(d)** only plants

14. Which of the following is a double displacement reaction? 1

(a) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(b)** $CaCO_3 \longrightarrow CaO + CO_2$ **(c)** $C + O_2 \longrightarrow CO_2$ **(d)** $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$

15. The pH of gastric juice is about: 1

(a) 7.4 **(b)** 5.5 **(c)** 10 **(d)** 1.4

16. The reactivity of sodium is explained by: 1

(a) its high melting point **(b)** its hardness
(c) its large mass **(d)** its easy loss of an electron

17. Soaps are sodium or potassium salts of: 1

(a) short-chain alcohols **(b)** long-chain carboxylic acids
(c) aldehydes **(d)** alkanes

18. Aerobic respiration in humans yields: 1

(a) only energy **(b)** lactic acid
(c) CO_2, water and energy **(d)** ethanol and CO_2

19. The movement of a shoot towards light is called: 1

(a) chemotropism **(b)** hydrotropism **(c)** geotropism **(d)** phototropism

20. The male gamete in a flowering plant is formed in the 1

(a) stigma **(b)** ovary **(c)** pollen grain **(d)** ovule

21. In a dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) seeds, the F1 seeds are 1

(a) round green **(b)** wrinkled yellow
(c) all round yellow **(d)** all wrinkled green

22. The refractive index of a medium is the ratio of the speed of light 1

(a) in glass to that in water only **(b)** in the medium to that in vacuum
(c) in air to that in water only **(d)** in vacuum to that in the medium

23. Myopia is corrected by using 1

(a) a concave lens **(b)** a plane mirror
(c) a convex lens **(d)** a cylindrical lens

24. Ohm's law states that the current through a conductor is 1

(a) independent of the potential difference
(b) inversely proportional to the potential difference
(c) proportional to the square of V
(d) directly proportional to the potential difference, at constant temperature

25. The field pattern of a solenoid is similar to that of 1

(a) a straight wire **(b)** a horseshoe magnet only
(c) a plane mirror **(d)** a bar magnet

26. A food chain always begins with 1

(a) carnivores **(b)** herbivores **(c)** producers **(d)** decomposers

27. $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ is a displacement reaction because: 1

(a) zinc is more reactive and displaces copper from its salt
(b) copper gets oxidised
(c) both metals are exchanged as anions
(d) zinc is less reactive than copper

28. Milk of magnesia, an antacid, is: 1

(a) magnesium hydroxide **(b)** magnesium carbonate
(c) magnesium sulphate **(d)** magnesium chloride

29. Metals like sodium and potassium are extracted from their ores by: 1

(a) reduction with carbon **(b)** roasting
(c) heating alone **(d)** electrolysis of molten chlorides

30. The property of carbon to form long chains by bonding with itself is called: 1

(a) catenation **(b)** isomerism **(c)** polymerisation **(d)** allotropy

31. The functional unit of the kidney is: 1

(a) neuron **(b)** nephron **(c)** alveolus **(d)** villus

32. The hormone responsible for the fight or flight response is: 1

(a) thyroxine **(b)** adrenaline **(c)** insulin **(d)** growth hormone

33. Sperms are produced in the 1

(a) ureter **(b)** urethra **(c)** testes **(d)** prostate

34. In human beings, the sex of the child is determined by 1

(a) the type of sperm that fertilises the egg
(b) the mother's diet
(c) the egg alone
(d) environmental temperature

35. A convex lens is also called 1

(a) a diverging lens **(b)** a cylindrical lens
(c) a plane lens **(d)** a converging lens

36. Splitting of white light into its seven colours is called 1

(a) dispersion **(b)** refraction **(c)** reflection **(d)** scattering

37. For resistors in series, the equivalent resistance is 1

(a) 1/R1 + 1/R2 **(b)** less than each
(c) R1 + R2 + R3 **(d)** the average of the resistances

38. Electromagnetic induction is the production of 1

(a) force on a coil
(b) induced current in a coil due to a changing magnetic field
(c) heat by current
(d) a magnetic field by a steady current

39. The ozone layer is present in the 1

(a) hydrosphere **(b)** troposphere **(c)** lithosphere **(d)** stratosphere

40. In the reaction $\mathrm{Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe}$, the type of reaction is: 1

(a) Combination **(b)** Decomposition
(c) Double displacement **(d)** Displacement

41. A solution reacts with crushed egg-shells to give a gas that turns lime water milky. The solution contains: 1

(a) HCl **(b)** NaOH **(c)** $\mathrm{Na_2SO_4}$ **(d)** NaCl

42. Which of the following methods is suitable for preventing an iron frying pan from rusting? 1

(a) applying paint **(b)** applying a coating of zinc
(c) applying grease **(d)** all of these

43. The number of covalent bonds in a molecule of pentane $\mathrm{C_5H_{12}}$ is: 1

(a) 14 **(b)** 16 **(c)** 15 **(d)** 17

44. The xylem in plants is responsible for: 1

(a) transport of food **(b)** transport of amino acids
(c) transport of oxygen **(d)** transport of water

45. The gap between two neurons is called: 1

(a) axon **(b)** myelin **(c)** dendrite **(d)** synapse

46. Multiple fission is seen in 1

(a) Amoeba **(b)** Planaria **(c)** Hydra **(d)** Plasmodium

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47. Mendel chose the garden pea for his experiments because 1

(a) it has many contrasting characters and is easy to grow
(b) it has no flowers
(c) it reproduces only asexually
(d) it grows only in cold regions

48. The image formed by a plane mirror is 1

(a) real and diminished
(b) virtual and magnified
(c) virtual, erect and of the same size as the object
(d) real and inverted

49. Most of the refraction of light entering the eye takes place at the 1

(a) retina **(b)** outer surface of the cornea
(c) eye lens **(d)** iris

50. The SI unit of electric current is 1

(a) coulomb **(b)** watt **(c)** volt **(d)** ampere

END OF QUESTION PAPER **Total: 50 Marks**

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